

TROLL STATION, Antarctica

WMO: 895040

Lat: 72.012S

Lon: 2.541E

Elev: 1277

StdP: 86.90

Time Zone: 0.00 (TRL)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) -37.4	(c) -35.9	(d) -44.9	(e) 0.1	(f) -35.6	(g) -43.4	(h) 0.1	(i) -33.8	(j) 23.8	(k) -14.9	(l) 21.1	(m) -17.0	(n) 1.4	(o) 230	(p) 1.265

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 1	(b) 5.6	(c) 0.2	(d) -4.4	(e) -1.6	(f) -5.6	(g) -2.8	(h) -6.3	(i) -5.1	(j) -2.0	(k) -5.8	(l) -2.8	(m) -6.5	(n) -3.7	(o) 3.9	(p) 20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
-9.2	2.0	-5.0	-10.4	1.8	-5.5	-11.5	1.6	-5.9	2.3	-1.8	1.2	-2.6	0.0	-3.6	-2.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.7	18.6	14.8	-39.5	1.3	2.1	1.2	-41.1	2.2	-42.3	2.9	-43.5	3.6	-45.1	4.5
			-39.6	-3.5	2.2	0.9	-41.2	-2.9	-42.5	-2.4	-43.7	-1.9	-45.3	-1.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-17.3	-6.8	-9.7	-15.6	-19.8	-21.1	-23.0	-25.3	-24.7	-24.0	-18.6	-12.0	-6.9		
	DBStd	8.09	2.38	3.38	3.83	5.28	5.74	5.01	5.82	6.03	5.66	5.07	3.19	2.26		
	HDD10.0	9973	522	552	794	893	963	990	1093	1076	1021	886	661	524		
	HDD18.3	13015	780	785	1052	1143	1221	1240	1351	1334	1271	1144	911	782		
	CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	3.9	4.1	4.1	3.2	3.8	4.3	3.5	3.3	4.0	3.5	4.2	4.8	4.6		
(15) Precipitation	PrecAvg	271	19	20	19	17	27	23	21	25	19	26	33	22		
	PrecMax	345	43	42	42	35	66	59	44	68	44	53	78	58		
	PrecMin	196	6	4	5	5	10	1	6	8	3	9	1	4		
	PrecStd	42	9	10	9	9	13	12	10	14	10	11	19	13		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.7	4.0	-4.0	-6.1	-6.4	-9.6	-9.7	-7.9	-9.2	-5.9	-2.0	1.9		
		MCWB	-3.0	-2.4	-7.3	-7.8	-9.0	-11.3	-11.5	-10.2	-11.8	-9.3	-6.7	-3.9		
	2%	DB	0.2	-0.2	-7.0	-9.0	-9.0	-11.9	-12.4	-10.8	-11.2	-7.8	-4.3	-0.7		
		MCWB	-4.4	-4.8	-9.7	-11.3	-11.1	-13.3	-13.8	-12.9	-13.2	-10.6	-7.8	-5.0		
	5%	DB	-1.6	-2.5	-8.5	-10.5	-11.0	-13.8	-14.7	-13.0	-13.5	-9.7	-5.9	-2.1		
		MCWB	-5.4	-6.4	-10.8	-12.7	-12.6	-15.3	-16.1	-14.7	-15.1	-12.0	-9.0	-6.0		
	10%	DB	-2.8	-4.6	-9.9	-11.9	-12.6	-15.5	-16.6	-15.6	-15.3	-11.3	-7.1	-3.2		
		MCWB	-6.1	-7.7	-11.9	-13.7	-14.1	-16.8	-17.8	-17.1	-16.8	-13.1	-9.9	-6.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-3.8	-5.2	-8.2	-7.2	-8.5	-11.1	-11.3	-9.8	-11.7	-9.0	-6.6	-4.0		
		MCDB	0.2	-0.9	-5.5	-6.0	-6.7	-9.9	-9.9	-8.0	-9.7	-6.3	-3.0	-0.9		
	2%	WB	-4.9	-6.6	-9.8	-11.0	-10.8	-13.2	-13.8	-12.8	-13.3	-10.6	-8.0	-5.2		
		MCDB	-1.7	-3.3	-7.7	-8.8	-9.4	-12.0	-12.4	-11.1	-11.4	-8.1	-4.8	-2.2		
	5%	WB	-5.5	-7.5	-10.9	-12.4	-12.5	-15.2	-16.0	-14.7	-15.0	-11.9	-9.1	-6.1		
		MCDB	-2.5	-4.9	-8.9	-10.5	-11.1	-13.9	-14.6	-12.9	-13.5	-9.8	-6.6	-3.0		
	10%	WB	-6.2	-8.4	-12.2	-13.5	-14.0	-16.7	-17.8	-17.0	-16.7	-13.2	-10.1	-6.7		
		MCDB	-3.4	-6.0	-10.3	-11.8	-12.7	-15.5	-16.6	-15.6	-15.2	-11.4	-7.7	-3.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.6	6.9	6.9	6.7	6.9	6.9	7.0	7.3	7.6	7.4	6.2	5.2		
		MCDBR	8.3	11.0	7.8	6.7	6.4	7.6	6.7	7.8	7.8	7.5	8.4	7.0		
	5% WB	MCWBR	3.8	4.9	4.8	5.3	5.3	6.7	5.9	6.8	6.6	6.0	5.4	3.8		
		MCWBR	5.5	6.1	5.9	6.4	6.0	7.2	6.6	7.7	7.2	7.2	6.8	5.5		
(40) Clear-Sky Solar Irradiance	taub	0.199	0.187	0.165	0.109	N/A	N/A	N/A	N/A	0.106	0.151	0.169	0.187	0.199		
		taud	2.292	2.302	2.271	1.973	N/A	N/A	N/A	1.942	2.208	2.251	2.271	2.276		
	Ebn at Noon	1057	1010	917	664	0	0	0	0	650	924	1029	1069	1073		
		Edh at Noon	103	84	59	25	0	0	0	25	60	88	104	110		
(44) All-Sky Solar Radiation	RadAvg	8.86	5.74	2.65	0.57	0.00	0.00	0.00	0.00	0.21	1.72	4.53	7.92	9.87		
	RadStd	0.18	0.18	0.10	0.03	0.00	0.00	0.00	0.00	0.01	0.08	0.14	0.28	0.18		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page